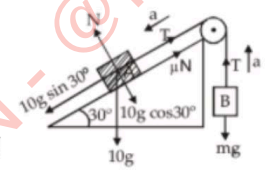
QPT-2 Solution

1. **Ans: (b)** 3.3



Solution: Considering the equilibrium of A, we get $\begin{aligned} 10\mathrm{a} &= 10\ \mathrm{g}\sin 30^\circ - \mathrm{T} - \mu\mathrm{N} \\ \text{where } \mathrm{N} &= 10\ \mathrm{g}\cos 30^\circ \end{aligned}$ $\therefore 10\mathrm{a} = \frac{10}{2}\ \mathrm{g} - \mathrm{T} - \mu \times 10\ \mathrm{g}\cos 30^\circ$ but $\mathrm{a} = 0,\ \mathrm{T} = \mathrm{m_B g}\ 0 = 5\ \mathrm{g} - \mathrm{m_B g} - \frac{0.2\sqrt{3}}{2} \times 10 \times \mathrm{g} \Rightarrow \mathrm{m_B} = 3.268 \approx 3.3\ \mathrm{kg}$

2. **Ans: (a)** force and displacement are perpendicular to each other

Solution: In a circular path, the centripetal force acts towards the centre of circle and the displacement is along the tangent to the circle. Hence, $\theta = 90^\circ$ $\therefore$ Work done $W = \mathbf{F} \cdot \mathrm{s} = Fs\cos\theta = Fs\cos 90^\circ = 0$

3. **Ans: (b)** $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$

Solution: $\mathrm{F_x} = -\frac{\partial \mathrm{U}}{\partial \mathrm{x}} = \sin(\mathrm{x} + \mathrm{y})$

$$\mathrm{F_y} = -\frac{\partial \mathrm{U}}{\partial \mathrm{x}} = \sin(\mathrm{x} + \mathrm{y})$$
$$\mathrm{F_x} = \sin(\mathrm{x} + \mathrm{y})]_{(0,\pi/4)} = \frac{1}{\sqrt{2}}$$
$$\mathrm{F_y} = \sin(\mathrm{x} + \mathrm{y})]_{(0,\pi/4)} = \frac{1}{\sqrt{2}}$$
$$\therefore \mathrm{F} = \frac{1}{\sqrt{2}}[\hat{\mathrm{i}} + \hat{\mathrm{j}}]$$

4. **Ans: (c)** 144 N

Solution: Velocity, $v = \frac{\mathrm{d}x}{\mathrm{d}t} = 3t^2 - 75$

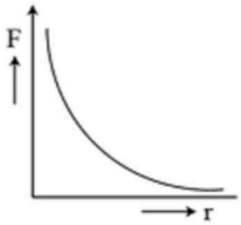
Acceleration, $a = \frac{\mathrm{d}v}{\mathrm{d}t} = 6t$

Force, $F = ma = 36t$

At, $t = 4$ s

$F = 144$ N

5. **Ans: (c)**



Solution: $F = ma = \frac{mv^2}{r} \Rightarrow F \propto \frac{1}{r} \Rightarrow$ Rectangular hyperbola.

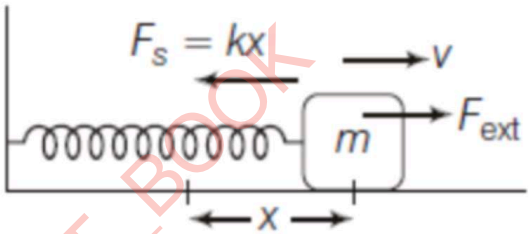
6. **Ans: (a)** 5 m

Solution: Given, $m = 5$ kg and $\mathrm{KE} = 490$ J By the law of conservation of energy

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mgh$$
$$490 = 245 + 5 \times 9.8 \times h$$
$$h = \frac{490 - 245}{5 \times 9.8}$$
$$h = \frac{245}{49}$$
$$= 5\ \mathrm{m}$$

7. **Ans: (c)** $W_{F_s} + W_{F_{ext}} > 0$

Solution: Spring-force, $F_s = kx$



At $x = \frac{mg}{k}$

$F_s = kx = k\frac{mg}{k} = mg = F_{\text{ext}}$

So, $F_s = F_{ext}$

But at this point, the block must have gain some KE as its motion was accelerated.

So, $W_{F_s} + W_{F_{ext}} = \Delta\mathrm{KE}$

[from work-energy theorem]

$W_{F_s} + W_{F_{ext}} > 0$

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8. **Ans: (d)** $32\ \mathrm{N}$

Solution: For equilibrium of all 3 masses, $a = \frac{T_3}{m_1+m_2+m_3}$ For equilibrium of m_1 & m_2 $T_2 = (m_1 + m_2) \cdot a$ or, $T_2 = \frac{(m_1+m_2)T_3}{m_1+m_2+m_3}$ Given $m_1 = 10\ \mathrm{kg},\ m_2 = 6\ \mathrm{kg},\ m_3 = 4\ \mathrm{kg}\ T_3 = 40\ \mathrm{N} \therefore T_2 = \frac{(10+6)\cdot 40}{10+6+4} = 32\ \mathrm{N}$

9. **Ans: (d)** $2\tan\phi$

Solution: Acceleration of block while sliding down upper half $= g\sin\phi$ retardation of block while sliding down lower half $= -(g\sin\phi - \mu g\cos\phi)$ For the block to come to rest at the bottom, acceleration in I half = retardation in II half. $g\sin\phi = -(g\sin\phi - \mu g\cos\phi) \Rightarrow \mu = 2\tan\phi$ Alternative method: According to work-energy theorem, $W = \Delta K = 0$ (Since initial and final speeds are zero) $\therefore$ Work done by friction + Work done by gravity $= 0$ i.e., $-(\mu mg\cos\phi)\frac{\ell}{2} + mg\ell\sin\phi = 0$ or $\frac{\mu}{2}\cos\phi = \sin\phi$ or $\mu = 2\tan\phi$

10. **Ans: (d)** Work performed and elapsed time of work

Solution: Power is defined as the rate of doing work. For the automobile, the power output is the amount of work done (overcoming friction) divided by the length of time in which the work was done.

11. **Ans: (a)** $\sqrt{\frac{2p}{a}}$

Solution:

Given, $F = at$

when, $t = 0$, then linear momentum $= p$

when, $t = T$,Â then linear momentum $= 2p$

According to Newton's law of motion, applied force, $F = \frac{dp}{dt}$

orÂ $dp = Fdt$

$dp = atdt$

orÂ â^«$_p^{2p} dp = a$â^«$_0^T tdt$

$(2p - p) = a$Â$\cdot \frac{T^2}{2}$

â‡’ $\frac{2p}{a} = T^2$â‡’$T = \sqrt{\frac{2p}{a}}$

.

12. **Ans: (c)** $10\ \mathrm{J}$

Solution: $W = Fs\cos\theta = 10 \times 2\cos 60^\circ = 10\ \mathrm{J}$,

13. **Ans: (a)** $\frac{\mu m}{M} g$

Solution: First we will assume that the both the block & plate will be moving together. If the common acceleration comes to out be greater than the maximum acceleration plate can have then we will treat both the block & plate separately.
Maximum Friction between plate and block: $\mu N = \mu mg$
Therefore maximum acceleration of plate is $\frac{\mu mg}{M}$
If moving together:
Acceleration: $a = \frac{F}{M_t} = \frac{3\mu mg}{M} + m$
As $a = \frac{3\mu mg}{M} + m > \frac{3\mu mg}{M} + 2M = \frac{\mu mg}{M}$
Therefore block and plate will move separately and the acceleration of plate will be equal to its maximum possible acceleration $\frac{\mu mg}{M}$,

14. **Ans: (b)** $1/2Ky(2x + y)$

Solution: In the string elastic force is conservative in nature. $\therefore W = -\Delta U$ Work done by elastic force of string,

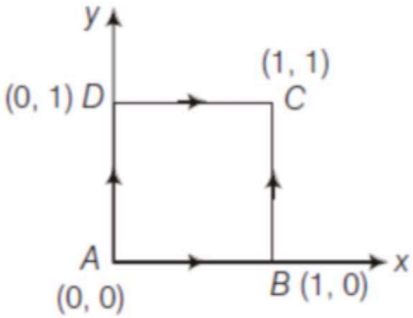
$$W = -(U_F - U_i) = U_i - U_F$$
$$W = \frac{1}{2}kx^2 - \frac{k}{2}(x+y)^2$$
$$= \frac{1}{2}kx^2 - \frac{1}{2}k\left(x^2 + y^2 + 2xy\right)$$
$$= \frac{1}{2}kx^2 - \frac{1}{2}ky^2 - \frac{1}{2}kx^2 - \frac{1}{2}k(2xy)$$
$$= -kxy - \frac{1}{2}ky^2$$

Therefore, the work done against elastic force $W_{external} = -W = \frac{Ky}{2}(2x + y)$;

15. **Ans: (a)** $(-1, 1)$

Solution: $W_{ABC} = W_{AB} + W_{BC}$
$= \int_{x=0}^{x=1}\left(y^2\hat{i} - x^2\hat{j}\right)\cdot dx\hat{i} + \int_{y=0}^{y=1}\left(y^2\hat{i} - x^2\hat{j}\right)\cdot dy\hat{j}$

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For $AB, y = 0$ and $BC, x = 1$

$W_{ABC} = \int_{X=0}^{X=1} \left(0\hat{i} - x^2\hat{j}\right) \cdot dx\hat{i} + \int_{y=0}^{y=1} \left(y^2\hat{i} - \hat{j}\right) \cdot dx\hat{j}$

$= 0 + (-1)\int_0^1 dy = -1$

Now, $W_{ADC} = W_{AD} + W_{DC}$

$= \int_{y=0}^{y=1} \left(y^2\hat{i} - x^2 j\right) \cdot dy\hat{j} + \int_{y=0}^{x=1} \left(y^2\hat{i} - x^2\hat{j}\right) \cdot dx\hat{i}$

For $AD, x = 0$ and for $DC, y = 1$

$\Rightarrow W_{ADC} = \int_0^1 \left(y^{2\hat{i}} - 0\right) dy\hat{j} + \int_0^1 \left(1\hat{i} - x^{2\hat{j}}\right) dx\hat{i}$

$= 0 + \int_0^1 dx = 1$

Hence, $(W_1, W_2) = (-1, 1)$!

16. **Ans: (a)** 3

Solution: Mass of particle, $\mathrm{m} = 500\ \mathrm{g} = 0.5\ \mathrm{kg}$

velocity of a particle, $\vec{v} = 2t\hat{i} + 3t^2\hat{j}$

$\vec{\mathrm{a}} = \frac{\mathrm{d}\vec{\mathrm{v}}}{\mathrm{dt}} = 2\hat{\mathrm{i}} + 6\mathrm{t}\hat{\mathrm{j}}$

at $\mathrm{t} = 1$, $\vec{\mathrm{a}} = 2\hat{\mathrm{i}} + 6\hat{\mathrm{j}}$

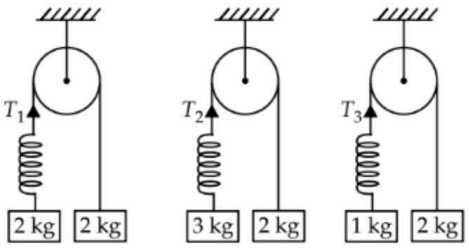
Force acting on the particle,

$\vec{\mathrm{F}} = \mathrm{ma} = 0.5(2\hat{\mathrm{i}} + 6\hat{\mathrm{j}}) = \hat{\mathrm{i}} + 3\hat{\mathrm{j}}$

$\vec{\mathrm{F}} = \hat{\mathrm{i}} + x\hat{\mathrm{j}}$ Hence $x = 3$ ^

17. **Ans: (d)** $x_2 > x_1 > x_3$

Solution:



If T_1, T_2, T_3 are the tensions in the strings in the three cases, we have

$$T_1 = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2 \times 2 \times 2g}{(2+2)} = 2g$$

$$T_2 = \frac{2 \times 3 \times 2g}{(3+2)} = 2.4g$$

$$T_3 = \frac{2 \times 1 \times 2g}{(1+2)} = 1.33g$$

As $x \propto T$ and $T_2 > T_1 > T_3$

$\therefore \quad x_2 > x_1 > x_3$

18. **Ans: (b)** 30

Solution: Given: - $\mu = 0.6$ - $g = 9.8\ \mathrm{m/s^2}$ - $r = 150\ \mathrm{m}$ Calculation:

$$v_{\min} = \sqrt{0.6 \cdot 9.8 \cdot 150}$$

$$v_{\min} = \sqrt{0.6 \cdot 1470}$$

$$v_{\min} = \sqrt{882}$$

$$v_{\min} \approx 29.7\ \mathrm{m/s}$$

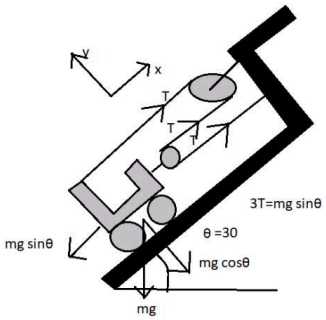
Conclusion: The minimum velocity is approximately $30\ \mathrm{m/s}$.

Therefore, the correct answer is: $30\ \mathrm{m/s}$. !

19. **Ans: (b)** $2.5\ \mathrm{ms}^{-2}$

Solution:

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Given,

A man pulls himself up the 30°incline by the method.

Here the man moving upward when force produced by tension in greater than vertical component of man,
$mg\sin\theta < 3T$
$mg\sin\theta + ma = 3T$
$a = \frac{3T - mg\sin\theta}{m}$

$= \frac{(3\times250)-(100\times10)\times\sin30^\circ}{100} = 2.5\ \text{m s}^{-2}$

;

20. **Ans: (b)** $3.6\ \text{J}$

Solution:
$$\text{Mass of the chain lying freely from the table } = \text{M}\frac{1}{\text{L}}$$
$$= 4\ \text{kg} \times \frac{0.6}{2}$$
$$= 1.2\ \text{kg}$$

The distance of center of mass of chain from the table $= \frac{1}{2} \times 0.6\ \text{m} = 0.3\ \text{m}$
Thus the work done in pulling the chain=mgh $= 1.2 \times 10 \times 0.3\ \text{J} = 3.6\ \text{J}$ *

21. **Ans: 6**

Solution: Since, $M >> m$, the velocity of M remains unchanged after m is placed on to it.
Acceleration of m is $\text{a} = \frac{\mu\text{mg}}{\text{m}} = \mu\text{g}$
Initial relative acceleration
$a_{mM} = a_{mE} - a_{ME} = a - 0 = a$
and initial relative velocity,
$\mu_{mM} = \mu_{mE} + \mu_{ME} = 0 - V = -v$
Hence.
$S_{mM} = \frac{v^2}{2\mu g} = \frac{(6)^2}{2\times0.3\times10} = 6\ \text{m}$

22. **Ans: 10**

Solution: Let, distance moved by A and B in any time interval as measured from natural length of spring are xA and xB respectively. Let, l is the natural length of spring, then elongation in spring at any instant is
$$\text{x} = (\text{x}_\text{A} - \text{x}_\text{B} - l)$$
$$\text{U} = \frac{\text{kx}^2}{2}$$
$$\frac{\text{dU}}{\text{dt}} = \text{P} = \frac{\text{k}}{2} \times 2\text{x}\frac{\text{dx}}{\text{dt}}$$
$$\Rightarrow \quad 20 = \text{kx}[\text{v}_\text{A} - \text{v}_\text{B}]$$
$$\Rightarrow \quad \text{kx} = \frac{20}{4-2} = 10\ \text{N}$$

23. **Ans: 20**

Solution: To calculate the work done in displacing the body from $x = 1\ \text{m}$ to $x = 5\ \text{m}$, we can use the concept that work done W is equal to the area under the force F vs. displacement x graph. Step-by-Step Calculation: 1. From $x = 1\ \text{m}$ to $x = 2\ \text{m}$: - The force $F = 10\ \text{N}$ is constant. - Displacement $\Delta x = 2 \quad 1 = 1\ \text{m}$. - Work done $W_1 = F \cdot \Delta x = 10 \times 1 = 10\ \text{J}$. 2. From $x = 2\ \text{m}$ to $x = 3\ \text{m}$: - The force $F = 5\ \text{N}$ is constant. - Displacement $\Delta x = 3 \quad 2 = 1\ \text{m}$. - Work done $W_2 = F \cdot \Delta x = 5 \times 1 = 5\ \text{J}$. 3. From $x = 3\ \text{m}$ to $x = 4\ \text{m}$: - The force $F = 5\ \text{N}$ (negative) is constant. - Displacement $\Delta x = 4 \quad 3 = 1\ \text{m}$. - Work done $W_3 = F \cdot \Delta x = 5 \times 1 = 5\ \text{J}$. 4. From $x = 4\ \text{m}$ to $x = 5\ \text{m}$: - The force $F = 10\ \text{N}$ is constant. - Displacement $\Delta x = 5 \quad 4 = 1\ \text{m}$. - Work done $W_4 = F \cdot \Delta x = 10 \times 1 = 10\ \text{J}$. Total Work Done: The total work done W_{total} from $x = 1\ \text{m}$ to $x = 5\ \text{m}$ is the sum of the individual works:
$W_{\text{total}} = W_1 + W_2 + W_3 + W_4$
$W_{\text{total}} = 10 + 5 \quad 5 + 10 = 20\ \text{J}$ Conclusion: The work done in displacing the body from $x = 1\ \text{m}$ to $x = 5\ \text{m}$ is 20 Joules.

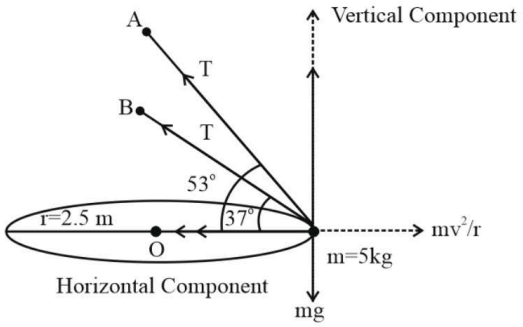
24. **Ans: 5**

Solution: Using work-energy theorem
$W_{mg} + W_F = \Delta KE$
$-mgh + Fd = \Delta KE$
$-1 \times 10 \times 4 + 18(5) = \Delta KE$
$\Delta KE = 50$
$\therefore n = 5$

25. **Ans: 5**

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Solution:



From the diagram in the condition of equilibrium, if T is the tension in the string, v is the velocity of motion, r is the radius of the loop then the horizontal component should be equal to centripetal force and vertical component should be equal to mg.

$T\cos 37^{\circ}+T\cos 53^{\circ}=\frac{mv^2}{r}$

$T\sin 53^{\circ}+T\sin 37^{o}=\frac{mv^2}{r}$

$\because \sin(90-\theta)=\cos\theta$

$T\sin 53^{\circ}+T\sin 37^{\circ}=mg$

$\Rightarrow \frac{mv^2}{r}=mg$

$\Rightarrow v=\sqrt{gr}$

$\Rightarrow v=\sqrt{10\times 2.5}=5\ \mathrm{m\ s^{-1}}$

$\Rightarrow \mathrm{v}=\sqrt{\mathrm{Rg}}=5\ \mathrm{m\ s^{-1}}$

26. **Ans: 20**

Solution: Given, mass of object $(m)=100$Â g

$=100$Ã—10^{-3}Â kg

Velocity of object $(v)=10\hat{i}$Â m/sÂ

Acceleration of object $(a)=\left(20\hat{i}+10\hat{j}\right)$Â m/s^2

We know that,

$\frac{d(KE)}{dt}=F.v=ma.v\left[\hat{a}\hat{}\mu K.E.=\frac{1}{2}mv^2\right]$

$=\left(100\text{Ã—}10^{-3}\right)\left(20\hat{i}+10\hat{j}\right).\left(10\hat{i}\right)$

$=100\text{Ã—}10^{-3}\text{Ã—}200$

$=\frac{100\text{Ã—}200}{1000}=20$Â kgÂ m^2/s^{3}Â

27. **Ans: 5**

Solution: First, find out the acceleration of A so for this

$\Rightarrow a=20-2F_2=20-2\times 8$

$a_A=4\ \mathrm{m\ s^{-2}}$

Now use pseudo concept (in which A is the non-inertial frame)

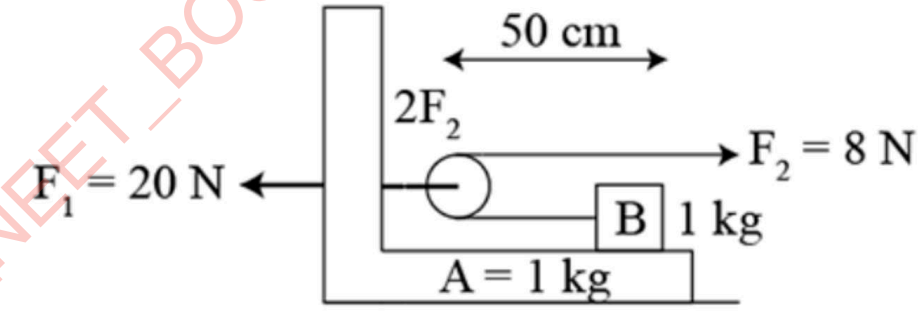
$\Rightarrow 8-4=4\ \mathrm{m\ s^{-2}}$

Now $\frac{50}{100}=\frac{1}{2}\times 4\times t^2$

$t=\frac{1}{2}=0.5\ \mathrm{s}$

$\Rightarrow 10t=5\ \mathrm{s}$





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28. **Ans: 32**

Solution: According to Newton's second law, the acceleration a of an object is given by: $a = \frac{F}{m}$ where F is the net force acting on the object and m is the mass of the object. Since the force F is constant, if the mass of the object increases to $4m$, the acceleration a' will be: $a' = \frac{F}{4m} = \frac{a}{4}$ The distance d traveled by an object accelerating from rest is given by: $d = \frac{1}{2}\,at^2$ For the object with mass m, the distance d traveled in time t is: $d = \frac{1}{2}\,at^2$ For the object with mass 4 m, the distance d traveled in time t' is: $d = \frac{1}{2}\,a'\left(t'\right)^2$ Substitute $a' = \frac{a}{4}$ into the second equation: $d = \frac{1}{2}\left(\frac{a}{4}\right)\left(t'\right)^2$ Set the distances equal to each other: $\frac{1}{2}\,at^2 = \frac{1}{8}\,a\left(t'\right)^2$ Divide both sides by $\frac{1}{8}\,a$: $4t^2 = \left(t'\right)^2$ Take the square root of both sides: $t' = 2t$ Given that $t = 16$ seconds, the time t' is: $t' = 2 \times 16 = 32$ seconds So, the object would take 32 seconds to reach the same final position if its mass was four times larger.

29. **Ans: 9**

Solution: Work done by the gun = total kinetic energy of the bullets

$$= n\,\frac{1}{2}\,mv^2 = 300 \times \frac{1}{2} \times 10 \times 10^{-3} \times (600)^2$$

$$= 150 \times 10 \times 10^{-3} \times 600 \times 600$$

$$\text{Power of gun} = \frac{\text{work done}}{\text{Time taken}}$$

$$= \frac{150 \times 10 \times 10^{-3} \times 600 \times 600}{60\ \text{s}}$$

$$= 9\ \text{kW} \quad (\because 1\ \text{minute} = 60\ \text{seconds})$$

30. **Ans: 1100**

Solution:

Friction force between A and B and between block B and surface S will oppose F

$$\hat{a}\hat{}\acute{}\hat{A}\ F = F_{AB} + F_{BS} = \hat{I}\text{¼}_{AB} M_A g + \hat{I}\text{¼}_{BS}(m_A + m_B)g$$

$$= 0.2\tilde{A}\text{—}100\tilde{A}\text{—}10 + 0.3(100 + 200)\tilde{A}\text{—}10$$

$$= 200 + 900 = 1100\hat{A}\ \text{N}$$

31. **Ans: (b)** $(\text{i}) < (\text{ii}) < (\text{iv}) < (\text{iii})$

Solution: The problem asks for the increasing order of the strength of hydrogen bonds in the given linkages. The strength of a hydrogen bond depends on the electronegativity of the atoms involved and the bond lengths. Let's analyze each case: 1. (i) $\text{O} - \text{H} \cdots \text{S}$: Sulfur is less electronegative than oxygen, so the hydrogen bond here will be relatively weak. 2. (ii) $\text{S} - \text{H} \cdots \text{O}$: Oxygen is more electronegative than sulfur, making this hydrogen bond stronger than in (i). 3. (iii) $\text{F} - \text{H} \cdots \text{F}$: Fluorine is the most electronegative element, and this involves a hydrogen bond with a negative fluoride ion, making it very strong. 4. (iv) $\text{F} - \text{H} \cdots \text{O}$: Fluorine is very electronegative, but oxygen is less so compared to fluorine, so this bond will be weaker than (iii) but stronger than the bonds involving sulfur. Based on this reasoning, the increasing order of the strength of the hydrogen bonds is: (i) < (ii) < (iv) < (iii)

32. **Ans: (b)** Perfect tetrahedral

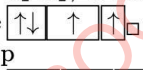
Solution: $\Rightarrow$ As DCl_3 is trigonal planar, hence, 'D' belongs to boron family.
$\Rightarrow \text{DCl}_3 + \text{ACl}_3 \rightarrow [\text{DCl}_2]^+[\text{ACl}_4]^-$ or $[\text{ACl}_2]^+[\text{DCl}_4]^-$
if anionic part is $[\text{ACl}_4]^-$ which has See - saw structure then cationic part will be $[\text{DCl}_2]^+$ where hybridization of D will be 'sp' and hence, shape will be linear.
$\Rightarrow$ If anionic part is DCl_4^- i.e, it is of tetrahedral shape the cation part will be ACl_2^+ which will be bent with sp^2 hybridisation.

33. **Ans: (a)** $\text{O}^{2-} > \text{F}^- > \text{O} > \text{F}$

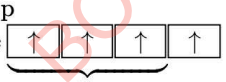
Solution: To determine the order of the radiu tor the species F^-, F^-, U, and U^-, we need to consider the following factors: 1. Electron Configuration and Effective Nuclear Charge: As the number of electrons increases or decreases, the radius of the ion changes due to the change in electron-electron repulsion and effective nuclear charge. 2. Anions vs. Cations: Anions (negatively charged ions) are larger than their parent atoms due to increased electron-electron repulsion. Cations (positively charged ions) are smaller due to reduced electron-electron repulsion and increased effective nuclear charge. 3. Comparing ions of Different Elements: Within the same period, anions are larger than neutral atoms, and the more negative the charge, the larger the anion. Within the same group, larger atoms generally have larger radii. Let's compare each species: 1. F^- (Fluoride ion): This ion has gained one electron compared to the neutral fluorine atom, leading to increased electron-electron repulsion and a larger radius than neutral fluorine. 2. O^{2-} (Oxide ion): This ion has gained two electrons compared to the neutral oxygen atom, leading to even more electron-electron repulsion and a larger radius than both neutral oxygen and F^-. 3. O (Neutral oxygen atom): The radius of neutral oxygen is smaller than the O^{2-} but larger than neutral fluorine due to fewer protons compared to fluorine, meaning the effective nuclear charge is less. 4. F (Neutral fluorine atom): This atom has the smallest radius among the given species due to its high effective nuclear charge and fewer electrons compared to the anions. Arranging these from largest to smallest radius, we get: 1. O^{2-} (largest radius due to most electron-electron repulsion) 2. F^- (next largest due to one extra electron compared to neutral fluorine) 3. O (neutral oxygen is smaller than F^- but larger than neutral fluorine) 4. F (smallest radius due to high effective nuclear charge) So the correct order of the radii is: $\text{O}^{2-} > \text{F}^- > \text{O} > \text{F}$

34. **Ans: (b)** $(\text{a})\hat{a}\check{Y}\P(\text{iii}), \hat{A}\ \hat{A}\ (\text{b})\hat{a}\check{Y}\P(\text{i}), \hat{A}\ \hat{A}\ (\text{c})\hat{a}\check{Y}\P(\text{iv}), \hat{A}\ (\text{d})\hat{a}\check{Y}\P(\text{ii})$

Solution: In $\text{C}_2\ \text{H}_2$, C undergoes, sp hybridisation.

Ground state 

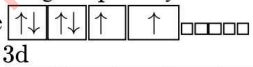
2s 2p

Excited state 

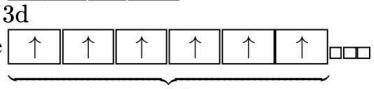
$\text{H} - \text{C} \equiv \text{C} - \text{H}$

sp

In SF_6, S undergoes sp^3d^2 hybridisation.

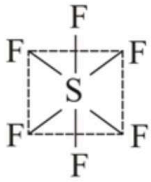
Ground state 

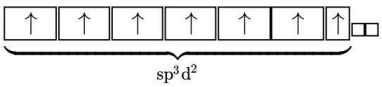
3s 3p 3d

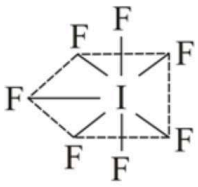
Excited state 

sp^3d^2

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In SO_2, S undergoes sp^2 hybridisation
In IF_7, I undergoes sp^3d^3 hybridisation.
Ground state 
3s 3p 3d
Excited state 



35. **Ans: (d)** decreases, increases

Solution: Bond order $= \frac{10-4}{2} = 3$
MOEC of N_2^+
$= \sigma 1s^2, \sigma^\star 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2$
$\equiv \pi 2p_y^2, \sigma 2p_z^1$
Bond order $= \frac{9-4}{2} = 2.5$
Here, on converting N_2 to N_2^+ bond order decreases, so bond strength decreases.
MOEC of $O_2 = \sigma 1s^2, \sigma^\star 1s^2, \sigma 2s^2, \sigma^\star 2s^2$
$\sigma p_z^2, \pi 2p_x^2 \equiv \pi 2p_y^2, \pi^* 2p_x^1 = \pi^* 2p_y^1$
Bond order $= \frac{10-6}{2} = 2$
MOEC of
$O_2^+ = \sigma 1s^2 \sigma^\star 1s^2, \sigma 2s^2 \sigma^* 2s^2, \sigma 2p_z^2$
$\pi 2p_x^2 \equiv \pi 2p_y^2, \pi^* 2p_x^1 \pi^* 2p_x^1$
Bond order $= \frac{10-5}{2} = 2.5$
On conversion of O_2 to O_2^+ bond order increases and hence bond strength increases.

36. **Ans: (a)** Many elements have several isotopes.

Solution:

Mendeleev's arranged all the known elements in increasing order of their atomic masses and gave the periodic law.

According to Mendeleev's Periodic law, The physical and chemical properties of the elements are the periodic function of their atomic masses.

Mendeleev designed a periodic table and arranged all the elements in increasing order of their atomic masses but not included the isotpes. As isotopes also have the different atomic masses. It is the major limitation of the Mendeleev's periodic table.

37. **Ans: (c)** $|$ IE of Process (ii) $| = |EA$ of Process (i) $|$

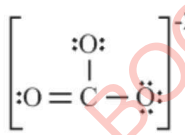
Solution: Amount of energy involved in loosing electron from outermost shell is known as IE while amount of energy involved in accepting one extra electron is known as EA and we know that
$IE = EA$
In process (ii), E is involved
$Cl^- \longrightarrow Cl + e^-$
While in process (i), EA is involved.
$Cl + e^- \longrightarrow Cl^-$
So, same amount of energy is released when electron is absorbed.
So, $|$ IE of process (ii) $| = |EA$ of process (i)$|$

38. **Ans: (a)** $4,\ 4,\ 3,\ 5.$

Solution:

Covalent bond is formed by sharing of electrons, which is generally observed between two non-metals.

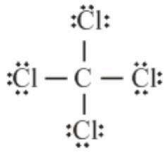
(a)



The number of covalent bonds in CO_3^{2-} is $= 4.$

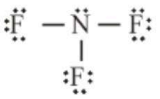
(b)

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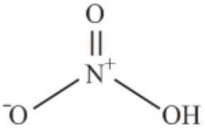
The number of covalent bonds in CCl_4 is $= 4$.

(c)



The number of covalent bonds in NF_3 is$= 3$.

(d)



The number of covalent bonds in HNO_3 is $= 5$.

39. **Ans: (b)** IE_1 of $Be < IE_1$ of B

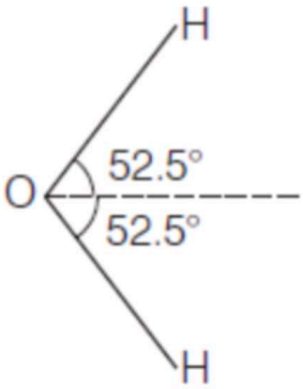
Solution: IE_1 of $Be > IE_1$ of B; in Be removal occurs from $2\,s$; In B it occurs from $2p$. More closer are subshell to nucleus more is IE.

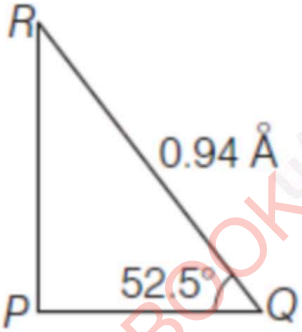
40. **Ans: (a)** (I), (II) and (IV)

Solution: (I) Be has smaller atomic radius compared to Mg. - True. Beryllium (Be) is in Period 2 and Magnesium (Mg) is in Period 3. As we move across a period from left to right, the atomic radius decreases due to increasing effective nuclear charge. Therefore, Be has a smaller atomic radius compared to Mg. (II) Be has higher ionization enthalpy than Al. - True. Ionization enthalpy generally decreases as we move down a group due to increased distance of the outer electrons from the nucleus and increased electron shielding. Since Be is in Period 2 and Al is in Period 3, Be has a higher ionization enthalpy compared to Al. (III) Charge/radius ratio of Be is greater than that of Al. - True. The charge/radius ratio is determined by the charge of the ion divided by its radius. Beryllium, with a +2 charge (as Be^{2-}) and a smaller radius compared to Aluminum (Al), will have a higher charge/radius ratio compared to Aluminum (Al^{3+}). Thus, Be has a greater charge/radius ratio than Al. (IV) Both Be and Al form mainly covalent compounds. - True. Both Beryllium and Aluminum tend to form covalent compounds due to their high charge densities which lead to significant polarization of the anions in their compounds. Beryllium forms covalent compounds due to its small size and high charge density, while Aluminum forms covalent compounds because it has a high charge and polarizing power. Based on this analysis, the correct statements are: - (I) Be has a smaller atomic radius compared to Mg. - (II) Be has higher ionization enthalpy than Al. - (III) Charge/radius ratio of Be is greater than that of Al. - (IV) Both Be and Al form mainly covalent compounds. The correct answer is: (I), (II) and (IV)

41. **Ans: (a)** 3.23×10^{-10}esu

Solution: Due to lone pair-lone pair repulsion the bond angle decreases to 105°





From the value of bond angle and vector moment, e can be calculated as

$PQ = QR\cos 52.5^\circ$

$= 0.94 \times 0.605 = 0.572\tilde{A}\ldots$

$= 0.572 \times 10^{-8}$ cm

We know that,

$\mu = e \times I = e \times PQ$

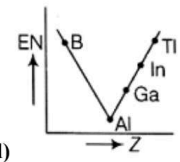
$e = \frac{\mu}{PQ} = \frac{1.85 \times 10^{-18}\text{esu cm}}{0.572 \times 10^{-8}\text{ cm}}$

$e = 3.23 \times 10^{-10}$esu

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42. **Ans: (a)** dipole-dipole interaction

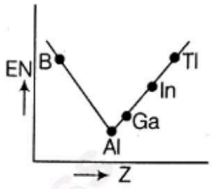
Solution: Due to dipole-dipole interaction the boiling point of alkyl halide is higher as compared to corresponding alkanes.



43. **Ans: (d)**

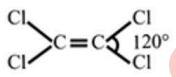
Solution: For group 13 elements, electronegativity decreases, from B to Al. The decrease in electronegativity from B to Al is due to increased size of Al. The remaining three elements are more electro negative than Al and their electronegativity increases from Ga to Tl. Thus, the correct order of electronegativity of group 13 elements are as follows.

Element	B	Al	In	Tl
EN (at Pauling scale)	2.0	1.5	1.6	1.7

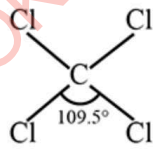


44. **Ans: (b)** 120° and $109 \cdot 5^{\circ}$

Solution:



1,1,2,2-tetrachloroethene
C is sp^2-hybridised.



tetrachloronemethane
C is sp^3-hybridised

45. **Ans: (d)** Both assertion and reason are true but the reason is not the correct explanation for the assertion

Solution: Let's analyze both the assertion and the reason: Assertion: For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group 7-9 elements. - True. The catalytic activity for hydrogenation reactions generally increases from Group 5 to Group 11 metals. Transition metals in Groups 7 to 9 (such as Ru, Rh, Pd, and Pt) are particularly effective catalysts for hydrogenation due to their ability to adsorb hydrogen and reactants efficiently, facilitating the reaction. Reason: The reactants are most strongly adsorbed on Group 7-9 elements. - False. While it is true that metals in Groups 7 to 9 are effective catalysts, the reason provided is not entirely correct. The key factor for their high catalytic activity is not just the strength of adsorption but also their optimal balance of adsorption and activation energies. For efficient hydrogenation, a metal catalyst needs to adsorb reactants effectively but not too strongly. Metals in Group 7-9 (like Pt, Pd, and Rh) have an optimal adsorption strength that allows them to efficiently facilitate the hydrogenation reaction. So, the correct evaluation is: - The assertion is true. - The reason is not a correct explanation for the assertion. Therefore, the correct option is: Both assertion and reason are true but the reason is not the correct explanation for the assertion.

46. **Ans: (a)** (X) < (Y) < (Z)

Solution: For third period, "s" block will form ionic oxide, 13th group forms amphoteric oxide, and 14th group element will form giant oxide (SiO_2).
(X) < (Y) < (Z)

47. **Ans: (b)** Bond length in NO is greater than that in NO^+

Solution:

Larger the magnitude of bond order, smaller will be the bond length.

48. **Ans: (a)** (a) -(iii, q), (b) -(i, r), (c) -(iv, p), (d) -(ii, s)

Solution:

	Column I		Column II		Column III
(a)	Bromine	(iii)	Liquid non-metal	(q)	$4s^2 4p^5$
(b)	Gold	(i)	Noble metal	(r)	Transition metal
(c)	Mercury	(iv)	Liquid metal	(p)	Amalgam
(d)	Iodine	(ii)	Crystalline non-metal	(s)	Violet

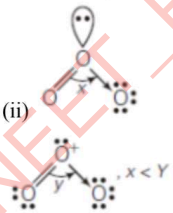
49. **Ans: (a)** Strength as Lewis acid $-BCl_3 > AlCl_3 > GaCl_3$

Solution: On moving down in group tendency to accept lone pair decreases. Therefore, lewis acid strength decreases $BCl_3 > AlCl_3 > GaCl_3$

50. **Ans: (b)** $AsH_3, O_3, NO_2^-, SOF_2$

Solution: (i) According to Drago rule, bond angle of NH_3, PH_3 and AsH_3 decreases top to bottom
$NH_3 > PH_3 > AsH_3$
Here, AsH_3 has smallest bond angle.

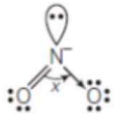
(ii)

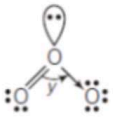


Lone pair-double bond repulsion in O_3 is greater than single electrondouble bond repulsion in O_3^+.
Therefore, the repulsion between Ione-pair single bond in O_3 is greater than repulsion between single electron single bond in O_3

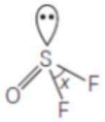
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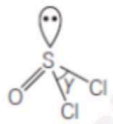
(iii) According to Bent rule, $\mathrm{N-O}$ bond in $\mathrm{NO_2^-}$ has more p-character than $\mathrm{O-O}$ bond in $\mathrm{O_3}$.





$\mathrm{O_3}$ has $\mathrm{sp^2}$ - hybridisation on central oxygen atom.
(iv) According to Bent rule, p-character of $\mathrm{S-Cl}$ bond in $\mathrm{SOCl_2}$ is lower than p-character of $\mathrm{S-F}$ bond in $\mathrm{SOF_2}$.





Hence, the correct choice regarding smallest bond angle is (b).

51. **Ans: 0**

Solution:



In $\mathrm{XeOF_4}$ no angle is of $90Â^\circ$ due to lp-bp repulsion.

52. **Ans: 6**

Solution:

The effective nuclear charge $\mathrm{Z_{eff}}$, is the net positive charge experienced by an electron in an atom.

The term "effective" is used because the shielding effect of negatively charged electrons prevents higher orbital electrons from experiencing the full nuclear charge of the nucleus due to the repelling effect of inner-layer electrons.

It increases across the period as the orbit remains the same but the number of protons increases as we move from left to right in the periodic table.

As we move down the group effective nuclear charge decreases or almost remains constant as an increase in shell number is balanced by the increase in the number of protons in the group.

Therefore, the order of $\mathrm{Z_{eff}}$ is:

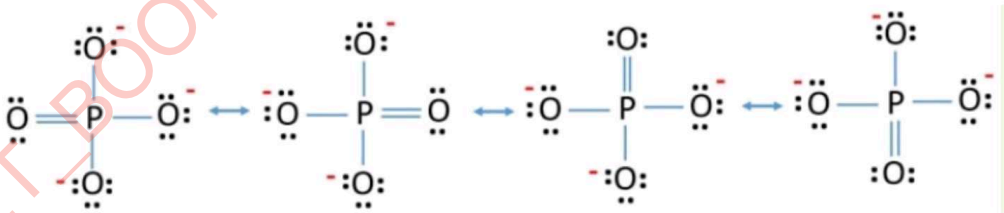
$$\mathrm{F > O > N > Cl > S > P > Si > Al > Mg > Na}$$

So, 6 elements have higher $\mathrm{Z_{eff}}$ than Silicon.

53. **Ans: 11**

Solution: 1. Calculate the Total Energy Required: - First Ionization Energy of Magnesium $\left(\mathrm{Mg \rightarrow Mg^+}\right): 7.646\mathrm{eV}$ - Second Ionization Energy of Magnesium $\left(\mathrm{Mg^+ \rightarrow Mg^{2+}}\right): 15.035\mathrm{eV}$ The total energy required to convert magnesium atoms to $\mathrm{Mg^{2+}}$ ions is the sum of the first and second ionization energies: $\text{Total energy } = 7.646\mathrm{eV} + 15.035\mathrm{eV} = 22.681\mathrm{eV}$ 2. Convert the Energy to $\mathrm{kJ/mol}$: - $1\mathrm{eV} = 96.485\ \mathrm{kJ/mol}$ Total energy in $\mathrm{kJ/mol} = 22.681\mathrm{eV} \times 96.485\ \mathrm{kJ/mol}$ per $\mathrm{eV} = 2196.468\ \mathrm{kJ/mol}$ 3. Determine the Number of Moles in 12 mg of Magnesium: - Molar Mass of Magnesium (Mg): $24.305\ \mathrm{g/mol}$ - Mass in grams: $12\mathrm{mg} = 0.012\ \mathrm{g}$ $\text{Number of moles } = \frac{0.012\ \mathrm{g}}{24.305\ \mathrm{g/mol}} = 4.94 \times 10^{-4}\ \mathrm{mol}$ 4. Calculate the Total Energy Needed: - Total energy for 1 mole of Mg to $\mathrm{Mg^{2+}}: 2196.468\ \mathrm{kJ}$ - Total energy for 0.000494 mol of Mg : $\text{Energy needed } = 2196.468\ \mathrm{kJ/mol} \times 4.94 \times 10^{-4}\ \mathrm{mol} = 1.084\ \mathrm{kJ}$ 5. Multiply by 10 and Round to the Nearest Integer: $1.084\ \mathrm{kJ} \times 10 = 10.84\ \mathrm{kJ}$ Rounding to the nearest integer gives 11. So, the amount of energy needed to convert all the atoms of magnesium in 12 mg of magnesium vapors into $\mathrm{Mg^{2+}}$ ions, when multiplied by 10 and rounded to the nearest integer, is 11 kJ.

54. **Ans: 2**



Solution: formal charge on each O -atom $= \frac{\text{Total charge}}{\text{No. of O-atom}} = \frac{-3}{4} = -0.75$ and $\mathrm{P-O}$ bond
Total No. of bonds $= \frac{5}{4} = 1.25$
Therefore $4A + 4B = 2$

55. **Ans: 4**

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Solution: $BeCl_2,\ N_3^-, N_2O, NO_2^+, O_3, SCl_2, ICl_2^-, I_3^-, XeF_2$

$BeC_2 \longrightarrow sp \longrightarrow$ linear

$N_3^- \longrightarrow sp \longrightarrow$ linear

$N_2O \longrightarrow sp \longrightarrow$ linear

$\overset{\oplus}{NO_2} \longrightarrow sp \longrightarrow$ linear

$O_3 \longrightarrow sp^2 \longrightarrow$ bent

$SCl_2 \longrightarrow sp^3 \longrightarrow$ bent

$I_3^- \longrightarrow sp^3\ d \longrightarrow$ linear

$ICl_2^- \longrightarrow sp^3\ d \longrightarrow$ linear

$XeF_2 \longrightarrow sp^3\ d \longrightarrow$ linear

So, among the following only four (4) has linear shape and no d-orbital is involved in hybridization.

56. **Ans: 4**

Solution: $Al,\ Ga,\ Tl$ and B belong to group 13 .

57. **Ans: 5**

Solution: $O_2^{2+}, CO, \overset{\grave{I}_{\ddot{}}}{C}N, NO^+, N_2$ have bond order 3 and are diamagnetic also.

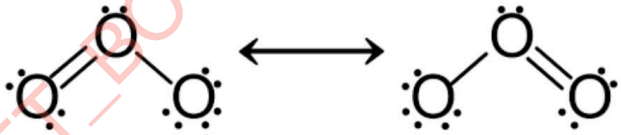
NO, O_2^+ have bond order 2.5 but are paramagnetic.

N_2^{2-}, O_2^{2-} have bond order 2 and 1 respectively.

58. **Ans: 6**

Solution:

We can have the following Lewis structure of the ozone molecule.



Therefore, there are six lone pairs on the molecule.

59. **Ans: 15**

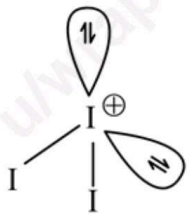
Solution: Its valence shell has 5 electrons $\left(ns^2, np^3\right)$ It belongs to 15 th group of the periodic table.

60. **Ans: 5**

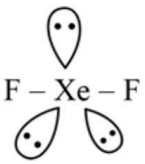
Solution:

At first, we will draw structures of given molecules.

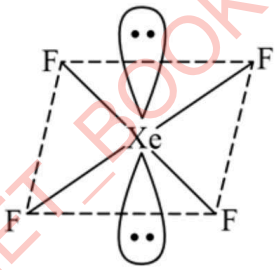
I_3^+ :



XeF_2 :

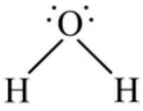


XeF_4 :

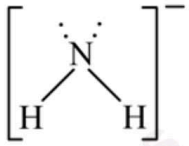


H_2O :

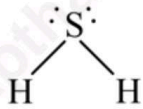
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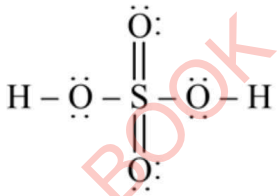
NH_2^- :



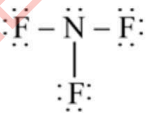
H_2S :



H_2SO_4 :



NF_3 :



From above structures of given compounds we can say that there are five molecules having two lone pairs on central atom.

61. **Ans: (a)** $-\frac{10}{31}$

Solution: Since, the first 11 terms in AP
$d = 2$
âˆ´ $a_{11} = a + 10d = a + 20$

The middle term of AP is $T_6 = a + 5d = a + 10$

For the next 11 terms in GP, $r = 2$
âˆ´ The middle term of GP is $b(2)^5$ where b is the first term of a GP which is the last term of AP.
âˆ´ $b(2)^5 = (a + 20)$Â 32

According to the given condition,
$a + 10 = (a + 20)$Â 32
â‡’ $31a = 10 - 640$Â Â Â Â â‡’Â Â Â Â Â $a =$Â $-\frac{630}{31}$
âˆ´ Middle term of entire sequence is 11th term.
âˆ´ $T_{11} =$Â $-\frac{630}{31} + 10$Ã—d
$=$Â $-\frac{630}{31} + 10$Â Ã—$2 =$Â $-\frac{10}{31}$

62. **Ans: (a)** 15°

Solution: $(\sin A + \sin B)^2 + (\cos A + \cos B)^2 = \frac{3}{2} + \frac{1}{2} = \frac{4}{2}$
$2 + 2(\sin A \sin B + \cos A \cos B) = 2$
$\cos(B - A) = 0 \Rightarrow B - A = \frac{\pi}{2} \Rightarrow B = \frac{\pi}{2} + A$
$\cos A + \cos B = \frac{1}{\sqrt{2}} \Rightarrow \cos A + \cos\left(\frac{\pi}{2} + A\right) = \frac{1}{\sqrt{2}}$
$\Rightarrow \cos A - \sin A = \frac{1}{\sqrt{2}} \Rightarrow \cos A \cdot \frac{1}{\sqrt{2}} - \sin A \frac{1}{\sqrt{2}} = \frac{1}{2}$
$\sin(45^o - A) = \frac{1}{2} = \sin 30^o \Rightarrow 45^o - A = 30^o$
$\Rightarrow A = 15^o$

63. **Ans: (c)** 2^{10}â‹…â‹…â‹…3^{12}

Solution:

Consider nine numbers $\frac{a}{2}, \frac{a}{2}, \frac{b}{3}, \frac{b}{3}, \frac{b}{3}, \frac{c}{4}, \frac{c}{4}, \frac{c}{4}, \frac{c}{4}$

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Using $\text{AM} \geq \text{GM}$, we get,

$\frac{\frac{a}{2}+\frac{a}{2}+\frac{b}{3}+\frac{b}{3}+\frac{b}{3}+\frac{c}{4}+\frac{c}{4}+\frac{c}{4}+\frac{c}{4}}{9} \geq \left(\left(\frac{a}{2}\right)^2\left(\frac{b}{3}\right)^3\left(\frac{c}{4}\right)^4\right)^{\frac{1}{9}}$

$\Rightarrow \frac{a+b+c}{9} \geq \left(\frac{a^2b^3c^4}{2^2 3^3 4^4}\right)^{\frac{1}{9}}$

$\Rightarrow \left(\frac{27}{9}\right)^9 \geq \frac{a^2b^3c^4}{2^2 3^3 2^8}$

$\Rightarrow a^2b^3c^4 \leq 2^{10} \cdot 3^{12}$

64. **Ans: (c)** 2

Solution:

Since,$2^1 = 2$,

$2^2 = 4$,

$2^3 = 8$,

$2^4 = 16$,

$2^5 = 32$

It is clear that unit place repeats after every fourth power, period of 4 is maintained.

Here, $2^{301} = \left(2^4\right)^{75} \cdot 2$

$= (16)^{75} \cdot 2$

$\therefore$ Digit at unit in $(16)^{75}$ is 6

$\therefore$ DIgit at unit place in 2^{301}

=(digit at units place in (16))2

= Unit digit in (16×2)

â€‹Hence, the remainder, when 2^{301} is divided by 5, is 2

65. **Ans: (a)** 62525

Solution: First term of each sets are $1, 2, 4, 7, \ldots$ Let $S = 1 + 2 + 4 + 7 + \ldots + T_n$

$S = 1 + 2 + 4 + \ldots + T_n$

On subtracting, we get

$0 = 1 + 1 + 2 + 3 + \ldots - T_n$

$T_n = 1 + (1 + 2 + 3 + \ldots (n-1) \text{ terms})$

$T_n = 1 + \frac{(n-1)n}{2}$ $\Rightarrow$

$T_{50} = 1 + \frac{49 \times 50}{2}$

$\therefore \quad T_{50} = 1226$

First term of 50th set is 1226 , therefore series is $1226, 1227, \ldots 50$ terms $S = \frac{50}{2}\left[2 \cdot 1226 + 49\right] = 62525$

66. **Ans: (b)** 135

Solution:

$A = G + \frac{3}{2}, G = H + \frac{6}{5}$

Using $G^2 = AH$ we get $G = 6,\ A = \frac{15}{2}$ and

$H = \frac{24}{5}$

Hence, $a + b = 15$ and $ab = 36$

a =12, b =3 or $a = 3,\ b = 12$

$\left|a^2 - b^2\right| = 135$

67. **Ans: (b)** 197

Solution: $\left(\sqrt{2}+1\right)^6 + \left(\sqrt{2}-1\right)^6 = 2\left({}^6C_0 2^3 + {}^6C_2 2^2 + {}^6C_4 2^1 + {}^6C_6 2^0\right)$

$\left(\sqrt{2}+1\right)^6 + \left(\sqrt{2}-1\right)^6 = 2(8 + 60 + 30 + 1) = 198$

$\left(\sqrt{2}+1\right)^6 = 198 - \left(\sqrt{2}-1\right)^6$

as $\sqrt{2} - 1$â‰ˆ$0.414 \Rightarrow \left(\sqrt{2}-1\right)^6$â€‹âˆˆ$(0,1)$

$\Rightarrow \left(\sqrt{2}+1\right)^6 = 198 - f, f$â€‹âˆˆ$(0,1)$

$\Rightarrow$ integral part of $\left(\sqrt{2}+1\right)^6 = 197$

68. **Ans: (b)** $b, \sqrt{a}, c$ are in geometric progression

Solution: $a = 1 + 10 + 10^2 + \ldots\ldots\ldots + 10^{54} = \frac{10^{55}-1}{9}$

$b = \frac{10^5-1}{9}$

$c = \frac{(10^5)^{11}-1}{10^5-1} = \frac{10^{55}-1}{10^5-1}$

$\Rightarrow \frac{a}{b} = c$

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$\Rightarrow a = bc$
$\Rightarrow b, \sqrt{a}, c$ are in G.P.

69. **Ans: (c)** $70\left(\frac{1-t}{1+t}\right)^4$

Solution:

$$\left[\left(t^{-1}-1\right)x+\left(t^{-1}+1\right)^{-1}x^{-1}\right]^8$$

$$=\left[\left(\frac{1}{t}-1\right)x+\left(\frac{1}{t}+1\right)^{-1}\frac{1}{x}\right]^8$$

Let T_{r+1} be the term independent of x, then

$$\mathrm{T}_{r+1} = {}^8\mathrm{C}_r\left(\tfrac{1}{t}-1\right)^{8-r}x^{8-r}\cdot\left(\tfrac{1}{t}+1\right)^{-r}\left(\tfrac{1}{x}\right)^r$$

$$= {}^8\mathrm{C}_r\left(\tfrac{1}{t}-1\right)^{8-r}\cdot\left(\tfrac{1}{t}+1\right)^{-r}\cdot x^{8-2r}$$

$\therefore 8-2r=0 \Rightarrow r=4$

$\therefore \mathrm{T}_5$ is the term independent of x and

$$\mathrm{T}_5 = {}^8C_4\left(\frac{1}{t}-1\right)^4\cdot\left(\frac{1}{t}+1\right)^{-4}$$

$$= {}^8\mathrm{C}_4\left(\frac{1-t}{t}\right)^4\cdot\left(\frac{1+t}{t}\right)^{-4}$$

$$= {}^8\mathrm{C}_4\left(\frac{1-t}{1+t}\right)^4 = \frac{8\cdot7\cdot6\cdot5}{4\cdot3\cdot2\cdot1}\left(\frac{1-t}{1+t}\right)^4$$

$$= 70\cdot\left(\frac{1-t}{1+t}\right)^4$$

70. **Ans: (b)** $1+\sin^2 1$

Solution: $f(0) = \cos^2(\cos\theta)+\sin^2(\sin\theta)$
Now, put $\theta = 0$
$\mathrm{f}(0) = \cos^2 1 + 0 = \cos^2 1$
$f(\pi/2) = 1+\sin^2 1$
Since, $\sin^2 1 > \cos^2 1$
$\left[\because \sin\theta > \cos\theta \forall \frac{\pi}{4} < \theta < \frac{\pi}{2} \ \& \ 1 > \frac{\pi}{4}\right]$
Now, seeing the options $1+\sin^2 1$ is greater than all other options.

71. **Ans: (d)** 5

Solution:

The remainder when 2021 divided by 7 is -2. Hence, the problem reduces to finding the remainder whenÂ $(-2)^{2023}$Â is divided by 7.

$= \frac{(-2)^{2022}\tilde{A}—(-2)}{7}$

$= -\frac{2\tilde{A}—2^{2022}}{7}$

$= \frac{-2\tilde{A}—\left(2^3\right)^{674}}{7}$

$= \frac{-2\tilde{A}—(8)^{674}}{7}$

$= \frac{-2\tilde{A}—(1+7)^{674}}{7}$

Using binomial theorem, we get

$= \frac{-2\tilde{A}—(1+7k)}{7}$Â whereÂ kÂ is an integer.

$= \frac{-2-14k}{7}$

Clearly, theÂ remainder whenÂ $-14k$Â is divisible byÂ 7Â isÂ 0.

$= \frac{-2}{7}$

$= \frac{-2-5+5}{7} = \frac{5}{7}$

Hence, the remainderÂ is 5.

72. **Ans: (d)** $\frac{11}{41}$

Solution: $\frac{\frac{p}{2}[2a_1+(p-1)d]}{\frac{q}{2}[2a_1+(q-1)d]} = \frac{p^2}{q^2}$

$\Rightarrow \frac{2a_1+(p-1)d}{2a_1+(q-1)d} = \frac{p}{q}$

$\frac{a_1+\left(\frac{p-1}{2}\right)d}{a_1+\left(\frac{q-1}{2}\right)d} = \frac{p}{q}$ For $\frac{a_6}{a_{21}}$, $\quad p=11, q=41$

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73. **Ans: (b)** $\frac{1}{3}$

Solution:

$\tan\theta = 3\tan\phi$
$\tan(\theta-\phi) = \frac{\tan\theta-\tan\phi}{1+\tan\theta\tan\phi}$
$= \frac{2\tan\phi}{1+3\tan^2\phi} = \frac{2}{\cot\phi+3\tan\phi}$
Using AM $\geq$ GM

$\frac{\cot\phi+3\tan\phi}{2} \geq \sqrt{3}$
$\Rightarrow \frac{2}{\cot\phi+3\tan\phi} \leq \frac{1}{\sqrt{3}}$
$\Rightarrow \tan(\theta-\phi) \leq \frac{1}{\sqrt{3}}$
$\Rightarrow \tan^2(\theta-\phi) \leq \frac{1}{3}$

74. **Ans: (a)** G.P.

Solution: $\frac{2}{\sqrt{c}+\sqrt{a}} = \frac{1}{\sqrt{b}+\sqrt{c}} + \frac{1}{\sqrt{a}+\sqrt{b}} = \frac{2\sqrt{b}+\sqrt{a}+\sqrt{c}}{(\sqrt{b}+\sqrt{c})(\sqrt{a}+\sqrt{b})}$
$\Rightarrow 2\sqrt{ab}+2b+2\sqrt{ac}+2\sqrt{bc} = 2\sqrt{bc}+2\sqrt{ac}+c+2\sqrt{ab}+a$
$\Rightarrow 2b = a+c$
$\therefore$ a, b, c, are in A.P.
$\Rightarrow$ ax, bx, cx, are in A.P.
$\Rightarrow ax+1, bx+1, cx+1$, are in A.P.
$\Rightarrow 9^{ax+1}, 9^{bx+1}, 9^{cx+1}$ are in G.P.

75. **Ans: (b)** ${}^{25}C_{20}$

Solution: Since there is no constant term the coefficient of 8th and 19th term are same as the binomial coefficients of 8th and 19th term.
${}^{n}C_7 = {}^{n}C_{18} \Rightarrow n = 7+18 = 25\ T_{n+1} = {}^{25}C_r \left(x^{\frac{4}{5}}\right)^{25-r} \left(r^{-\frac{1}{5}}\right)^{r} {}^{25}C_r\ x^{\frac{4}{5}(25-r)-\frac{r}{5}}$ To be independent of x, $\frac{100-5r}{5} = 0 \Rightarrow r = 20$ Hence, the required term is ${}^{25}C_{20}$

76. **Ans: (b)** $\mathrm{n} + \frac{4^{-\mathrm{n}}}{3} - \frac{1}{3}$

Solution: Consider $\frac{3}{4} + \frac{15}{16} + \frac{63}{64} + \ldots$ upto n terms $= \frac{2^2-1}{2^2} + \frac{2^4-1}{2^4} + \frac{2^6-1}{2^6}$ upto n term $= \left(1-\frac{1}{2^2}\right) + \left(1-\frac{1}{2^4}\right) + \left(1-\frac{1}{2^6}\right) + \ldots$ upto n terms
$= (1+1+1+\ldots.$ upto n terms$) - \left(\frac{1}{2^2}+\frac{1}{2^4}+\frac{1}{2^6}+\ldots \text{upto n terms}\right) = \mathrm{n} - \frac{1}{2^2}\left[\frac{1-\left(\frac{1}{2^2}\right)^{\mathrm{n}}}{1\frac{1}{2^2}}\right] = \mathrm{n} - \frac{1}{3}\left(1-4^{-\mathrm{n}}\right) = \mathrm{n} + \frac{4^{-\mathrm{n}}}{3} - \frac{1}{3}$

77. **Ans: (b)** 1214

Solution:

Given that, $f(x) = x^3 + ax^2 + bx + c$

Now, since the roots of the polynomial are in Arithmetic Progression,

Let the roots are $s-t,\ s,\ s+t$

$\Rightarrow s-t+s+s+t = \frac{-a}{1}$

$\Rightarrow -3s = a$

Hence, a must be multiple of 3.

From given options we see that only 1214 is not multiple of 3.

78. **Ans: (a)** $\mathrm{A}^2 - \mathrm{B}^2$

Solution: $(x+a)^n = {}^{n}C_0x^n + {}^{n}C_1x^{n-1}a + {}^{n}C_2x^{n-2}a^2 + {}^{n}C_3x^{n-3}a^3 + {}^{n}C_4x^{n-4}a^4 + \ldots.$
$= \left({}^{n}C_0x^n + {}^{n}C_2x^{n-2}a^2 + {}^{n}C_4x^{n-4}a^4 + \ldots.\right) + + \left({}^{n}C_1x^{n-1}a + {}^{n}C_3x^{n-3}a^3 + {}^{n}C_5x^{n-5}a^5\right) + \ldots. = A+B$ ....(i) Similarly, $(x-a)^n = A-B$ ....(ii) Multiplying eqns. (1) and (2), we get $\left(\mathrm{x}^2-\mathrm{a}^2\right)^{\mathrm{n}} = \mathrm{A}^2 - \mathrm{B}^2$

79. **Ans: (a)** $2^{211} - 2^{190}$

Solution:

$(x+1)(2x+1)\left(2^2x+1\right)\left(2^3x+1\right)\ldots\left(2^{20}x+1\right)$

$= 1\cdot 2\cdot 2^2\cdot 2^3\ldots 2^{20}$

$(x+1)\left(x+\frac{1}{2}\right)\left(x+\frac{1}{2^2}\right)\left(x+\frac{1}{2^3}\right)\cdots\left(x+\frac{1}{2^{20}}\right)$

$= 2^{\frac{20\times 21}{2}}(x+1)\left(x+\frac{1}{2}\right)\left(x+\frac{1}{2^2}\right)\ldots\left(x+\frac{1}{2^{20}}\right)$

Coefficient of $x^{20} = 2^{210}\left(1+\frac{1}{2}+\frac{1}{2^2}+\ldots+\frac{1}{2^{20}}\right)$

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$= \left(2^{21}\right)^{10}\left(\frac{1-\frac{1}{2^{21}}}{1-\frac{1}{2}}\right) = \left(2^{21}\right)^{9} 2\left(2^{21}-1\right) = 2^{211} - 2^{190}$

80. **Ans: (b)** independent of Î,

Solution: $2\sin^2â¡Î, + 2\cosâ¡Î,\left(Î, + Î±\right)\left(\cosâ¡\left(Î, - Î±\right) - \cosâ¡\left(Î, + Î±\right)\right) + \left(2\cos^2â¡Î,\left(Î, + Î±\right) - 1\right) = 2\sin^2â¡Î, + 2\left(\cos^2â¡Î, - \sin^2â¡Î±\right) - 1$
$= 2 - 2\sin^2â¡Î± - 1 = \cos^2â¡Î± - \sin^2â¡Î±$

81. **Ans: 3954**

Solution: Using the formula for the sum of a geometric progression, we find
$(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots + x^{1000}$
$= \frac{(1+x)^{1000}\left[1-\left(\frac{x}{1+x}\right)^{1001}\right]}{\left[1-\left(\frac{x}{1+x}\right)\right]} = \frac{(1+x)^{1000} - \frac{x^{1001}}{(1+x)}}{\frac{x+1-x}{(1+x)}} = (1+x)^{1001} - x^{1001}$
Hence, the coefficient of $x^{50} = {}^{1001}C_{50} = \frac{1001!}{50!951!}$
$\lambda = 1001, \mu = 50$ and $\mathrm{v} = 951$
$\lambda + 2\mu + 3v = 1001 + 100 + 2853 = 3954$

82. **Ans: 384**

Solution:

Let, the last element of set A is, $t_{200} = 2 + (200-1)2 = 400$

and the last element of set B is $t_{250} = 3 + (250-1)3 = 750$

Common elements are $\rightarrow 6, 12, 18 \ldots$

We have to find the common elements upto 400
$t_n = 6 + (n-1)6 \Rightarrow 400 = 6 + 6n - 6 \Rightarrow n = \frac{400}{6} = 66.6\ldots$
$t_{66} = 6 + (66-1)6 = 396$
Number of common elements $= 66$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 200 + 250 - 66 = 384 \Rightarrow k = 384$

83. **Ans: 2**

Solution: $200 < {}^{\mathrm{n}}\mathrm{C}_0 + {}^{\mathrm{n}}\mathrm{C}_1 + â€¦ + {}^{\mathrm{n}}\mathrm{C}_{\mathrm{n}} < 400$
â‡’ $2^{\mathrm{n}} < 400$
â‡’ $\mathrm{n} = 8$
$\mathrm{T}_{\mathrm{r}+1} = {}^{8}\mathrm{C}_{\mathrm{r}}\left(\sqrt[4]{\mathrm{x}^{-3}}\right)^{8-\mathrm{r}}\left(\mathrm{a}\sqrt[4]{\mathrm{x}^{5}}\right)^{\mathrm{r}}$
â‡’ $\mathrm{T}_{\mathrm{r}+1} = {}^{8}\mathrm{C}_{\mathrm{r}}\,\mathrm{a}^{\mathrm{r}}\mathrm{x}^{2\mathrm{r}-6}$
For this term to be independent of x,
$2\mathrm{r} - 6 = 0 â‡’ \mathrm{r} = 3$
$\mathrm{T}_4 = {}^{8}\mathrm{C}_3\mathrm{a}^3 â‡’ 448 = 56\mathrm{a}^3 â‡’ \mathrm{a}^3 = 8$
â‡’ $\mathrm{a} = 2$

84. **Ans: 5**

Solution:
$$(720+1)^{720} = {}^{720}C_0(720)^{720} + {}^{720}C_1(720)^{719} + \ldots + {}^{720}C_{718}(720)^2 + {}^{720}C_{719}(720) + {}^{720}C_{720}$$
$$= 1000k + \frac{(720)(719)}{2}(720)^2 + (720)(720) + 1$$
$$= 1000k' + 518400 + 1$$
$$\Rightarrow a = 1, \quad b = 0, c = 4$$

85. **Ans: 2**

Solution:

Let, $E = \sin\frac{\pi}{18} \cdot \sin\frac{5\pi}{18} \cdot \sin\frac{7\pi}{18}$

Now

$\sin\frac{7\pi}{18} = \sin\left(\frac{\pi}{2} - \frac{2\pi}{18}\right) = \cos\frac{2\pi}{18}$ and

$\sin\frac{5\pi}{18} = \sin\left(\frac{\pi}{2} - \frac{4\pi}{18}\right) = \cos\frac{4\pi}{18}$

So, $E = \sin\frac{\pi}{18} \cdot \cos\frac{2\pi}{18} \cdot \cos\frac{4\pi}{18}$

Multiply & divide by $2\cos\frac{\pi}{18}$

$E = \frac{2\cos\frac{\pi}{18}\sin\frac{\pi}{18} \cdot \cos\frac{2\pi}{18} \cdot \cos\frac{4\pi}{18}}{2\cos\frac{\pi}{18}}$

Now using $\cos A \cos 2A \cos 2^2 A \ldots\ldots \cos 2^{n-1}A = \frac{\sin 2^n A}{2^n \sin A}$, we can write

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$E=\frac{\sin\frac{8\pi}{18}}{8\cos\frac{\pi}{18}}=\frac{\sin\left(\frac{\pi}{2}-\frac{\pi}{18}\right)}{8\cos\frac{\pi}{18}}=\frac{1}{8}$=k

so $16k=2$

86. **Ans: 4**

Solution: Expression $=(1-x)^5\cdot(1+x)^4(1+x^2)^4$ $=(1-x)(1-x^2)^4(1+x^2)^4$ $=(1-x)(1-x^4)^4$ $\therefore$ Coefficient of $x^{13}={}^4C_3(-1)^3=4$

87. **Ans: 9**

Solution:

We have,

$\sum_{k=0}^{20}\left(\frac{A_k}{\alpha+k}\right)=\frac{1}{\alpha(\alpha+1).....(\alpha+20)}$

$\Rightarrow\frac{A_0}{\alpha}+\frac{A_1}{\alpha+1}+...+\frac{A_{20}}{\alpha+20}=\frac{1}{\alpha(\alpha+1).....(\alpha+20)}$

$\Rightarrow\frac{A_0(\alpha+1)(\alpha+2).....(\alpha+19)(\alpha+20)+A_2\alpha(\alpha+2).....(\alpha+19)(\alpha+20)+...+A_{20}\alpha(\alpha+1)(\alpha+2).....(\alpha+19)}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)}=\frac{1}{\alpha(\alpha+1)(\alpha+2).....(\alpha+20)}$

$\Rightarrow A_0[(\alpha+1)(\alpha+2).....(\alpha+19)(\alpha+20)]+A_2[\alpha(\alpha+2).....(\alpha+19)(\alpha+20)]+...+A_{20}[\alpha(\alpha+1)(\alpha+2).....(\alpha+19)]=1\quad....(i)$

Hence, coefficient of A_{13} is

$\alpha(\alpha+1)\dots(\alpha+12)(\alpha+14)\dots(\alpha+19)(\alpha+20).$

Similarly, coefficient of A_{14} is

$\alpha(\alpha+1)\dots(\alpha+13)(\alpha+15)\dots(\alpha+19)(\alpha+20)$

And, coefficient of A_{15} is

$\alpha(\alpha+1)\dots(\alpha+14)(\alpha+16)\dots(\alpha+19)(\alpha+20)$

Then, to find A_{14}, put $\alpha=-14$ in (i), then we get

$A_{14}=\frac{1}{(-14)(-13).....(-1)(1).....(6)}=\frac{1}{14!\cdot6!}$

Similarly, putting $\alpha=-15$, we get

$A_{15}=\frac{1}{(-15)(-14).....(-1)(1).....(5)}=-\frac{1}{15!\cdot5!}$

And, putting $\alpha=-13$, we get

$A_{13}=\frac{1}{(-13).....(-1)(1).....(7)}=\frac{-1}{13!\cdot7!}$

Hence,

$\frac{A_{14}}{A_{13}}=\frac{1}{14!\cdot6!}\times(-13!\times7!)=\frac{-7}{14}=-\frac{1}{2}$

$\frac{A_{15}}{A_{13}}=\left(-\frac{1}{15!\times5!}\right)\times(-13!\times7!)=\frac{42}{15\times14}=\frac{1}{5}$

Then,

$100\left(\frac{A_{14}}{A_{13}}+\frac{A_{15}}{A_{13}}\right)^2=100\left(-\frac{1}{2}+\frac{1}{5}\right)^2=9$

88. **Ans: 312**

Solution: $(x+2)^3(x+3)^2(x+5)^3=x^8+x^7(2+2+2+3+3+5+5+5)+x^6K+.....$
Clearly, K is equal to the sum taken two at a time of $2,2,2,3,3,5,5,5$.
$\Rightarrow K=\frac{(2+2+2+3+3+5+5+5)^2-(2^2+2^2+2^2+3^2+3^2+5^2+5^2+5^2)}{2}$
$=\frac{27^2-(4+4+4+9+9+25+25+25)}{2}$
$=\frac{729-105}{2}=\frac{624}{2}=312$=k

89. **Ans: 21**

Solution: To solve this problem, we need to find the values of p, q, and r such that: $(1+3+5+\cdots+p)+(1+3+5+\cdots+q)=1+3+5+\cdots+r$ Step 1: Find the Sum of Consecutive Odd Integers The sum of the first n consecutive odd integers is given by: $S_n=1+3+5+\cdots+(2n-1)=n^2$ So, the sum of the first k odd integers is k^2. Step 2: Express the Given Sums Let n be the number of terms in the sequence ending in p. Thus: $1+3+5+\cdots+p=n^2$ where $p=2n-1$. Similarly, let m be the number of terms in the sequence ending in q. Thus: $1+3+5+\cdots+q=m^2$ where $q=2m-1$. And: $1+3+5+\cdots+r=k^2$ where $r=2k-1$. Step 3: Solve the Equation The equation given is: $n^2+m^2=k^2$ This is a Pythagorean triple problem where n^2 and m^2 must sum to k^2. We need to find the smallest integers n, m, and k that satisfy this. Step 4: Find the Smallest Pythagorean Triple The smallest Pythagorean triple is $(3,4,5)$, where: $3^2+4^2=5^2$ So: $n=3,\quad m=4,\quad k=5$ Step 5: Calculate p, q, and r For $n=3: p=2n-1=2\times3-1=5$ For $m=4:$ $q=2m-1=2\times4-1=7$ For $k=5: r=2k-1=2\times5-1=9$ Step 6: Calculate $p+q+r$ $p+q+r=5+7+9=21$ Thus, the smallest possible value of $p+q+r$ is 21.

90. **Ans: 7**

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Solution: Let, $\tan 20^\circ = k$

then, $\tan 60^\circ = \frac{3\tan 20^\circ - \tan^3 20^\circ}{1 - 3\tan^2 20^\circ}$

$\sqrt{3} = \frac{3k - k^3}{1 - 3k^2}$

$\left(\sqrt{3}\left(1 - 3k^2\right)\right)^2 = \left(3k - k^3\right)^2$

$3\left(1 + 9k^4 - 6k^2\right) = 9k^2 + k^6 - 6k^4$

$3 + 27k^4 - 18k^2 = 9k^2 + k^6 - 6k^4$

$k^6 - 33k^4 + 27k^2 - 3 = 0$

$k^6 - 33k^4 + 27k^2 + 4 = 7$

Hence, $\tan^6 20^\circ - 33\tan^4 20^\circ + 27\tan^2 20^\circ + 4 = 7$

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